Herrings and relatives

PHYLUM	Chordata
CLASS	Osteichthyes
SUBCLASS	Actinopterygii
SUPERORDER	Clupeomorpha
ORDERS	1 (Clupeiformes)
FAMILIES	5
SPECIES	405

These small, streamlined fishes are widespread and abundant. Often forming vast schools, they are an important part of marine food chains and are vital to many coastal economies. The herring family includes sprats, shads, and pilchards; it also includes sardines, although this term is often used to refer to the young or adults of other species. Herrings and anchovies occur from the tropics to temperate zones, mostly in coastal marine waters with some in freshwater streams and lakes.

Anatomy

Members of the herring family are similar to one another in overall appearance. They have streamlined bodies that are flattened from side to side, with large scales and well-developed fins without any supporting spines. Most are silver with green or blue hues on the back, becoming progressively lighter on their undersides. Anchovies are generally longer and thinner than herrings, and have a large mouth with a prominent snout. In both anchovies and herrings, the swim bladder is linked to the ear apparatus, which is thought to enhance their hearing. At one time, herrings and their relatives were considered to be related to tarpons and eels because of similarities between their larvae, but now the consensus is that they are sufficiently different to form a group of their own.

CONSERVATION

Herrings have a long history of human exploitation. Easy to catch and to preserve, they are an important source of food and oil. During the days of sail, catches were relatively stable, but the herring harvest soared in the first half of the 20th century, with the introduction of powered fishing vessels. In the 1970s, the Atlantic herring population crashed. Some stocks are now sustainable and well-managed, but others are not.

Feeding

Almost all herrings and their relatives feed on planktonic organisms (mainly crustaceans and their larvae), which they filter out of the water by opening their large mouths to expose their gill rakers. Following the daily vertical movements made by plankton, they usually feed near the surface at night and move to lower depths during the day. Their feeding habits often vary seasonally; herrings and some other species cease to feed during the breeding season. Most members of this group feed in dense groups of tens of thousands of fishes, thus attracting predators, including other fishes (such as salmon and tuna), seabirds, and marine mammals. By contributing to the diet of so many other species, herrings and anchovies form a key element in many marine ecosystems.

SCHOOLING
Herring schools are at their largest during the breeding season, which usually falls in the warmer months of the year. Each female releases up to 60,000 eggs, which form a sticky mat on the seabed.

Clupea harengus

Atlantic herring

Location North Atlantic, North Sea, Baltic Sea

Length Up to 16 in (40 cm)
Weight Up to 25 oz (700 g)
Sex Male/Female
Status Least concern

For centuries, the sleek, silver-scaled herring has been one of the most important commercially fished species in the North Atlantic. Its numbers declined precipitously during the 20th century, due to improved fishing techniques, but are now recovering slowly because of active management. Like most of its relatives, the Atlantic herring is a plankton-feeder, and lives in large, highly mobile schools. It comes to the surface at night, but spends the day in deeper water. Across its wide range, the species is divided into many local races, which differ from each other in habits and in size. Each race uses a number of traditional spawning grounds. Young herrings, which initially resemble tiny eels, hatch on the seabed, but soon swim upward to feed close to the surface. They are eaten in vast numbers by other fishes, and only a tiny proportion of them survive to become adults.